

Quanyan Zhu

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ScholarGPS: scholargps.com/scholars/54961215509101/quanyan-zhu

Research.com: research.com/u/quanyan-zhu

Google Scholar: scholar.google.com/citations?user=Qt2WDh0AAAAJ

Abridged CV: selected activities since 2022. Full CV available upon request.

Employment

- Associate Professor, Department of Electrical and Computer Engineering, New York University, NY, USA, Sept. 2019–Present
- Affiliated Associate Professor, Center for Urban Science And Progress (CUSP), New York University, NY, USA, Sept. 2019–Present
- Assistant Professor, Department of Electrical and Computer Engineering, New York University, NY, USA, Jan. 2014–Aug. 2019
- Affiliated Assistant Professor, Center for Urban Science And Progress (CUSP), New York University, NY, USA, April 2019–Aug. 2019
- Postdoctoral Research Associate, Department of Electrical Engineering, Princeton University, NJ, USA, Sept. 2013–Dec. 2013 (Host: Vincent Poor)

Research Impact and Scholarly Metrics

- ScholarGPS® ID: 54961215509101. Highly Ranked Scholar – Lifetime: #11 in Game Theory and #12 in Physical Systems; Highly Ranked Scholar – Prior Five Years: #2 in Game Theory.
- Research.com Computer Science rankings: Current World Ranking #4399 and Current National Ranking #2046.
- Google Scholar citation metrics: 17,466 citations (10,149 since 2021); h-index 65 (48 since 2021); i10-index 283 (233 since 2021). Retrieved August 10, 2026.
- Internationally recognized researcher in game theory, cybersecurity, cyber-physical systems, AI security, resilient autonomy, and network science.
- Research contributions span strategic learning, control theory, cyber deception, federated learning, resilient AI, and multi-agent systems; developed foundational cyber-resilience frameworks integrating game theory, control theory, and learning theory.
- Known for influential work on defensive deception, resilient control, moving target defense, and AI-enabled cyber resilience.

Research Interests

- Dynamic Games and Control
- Multi-Agent Systems and Reinforcement Learning
- Human-Machine Interactions
- Cyber Resiliency
- Distributed Decision-Making and Mechanism Design
- Cyber-Physical-Human Systems
- Critical Infrastructure Protection

- Internet of Things
- Cyber Manufacturing
- Security of 5G Networks
- STEM Education

Education

- Doctor of Philosophy, Department of Electrical and Computer Engineering, University of Illinois at Urbana-Champaign, USA, Aug. 2008–May 2013. Ph.D. Dissertation: Game-Theoretic Methods for Security and Resilience of Cyber-Physical Systems, advised by Prof. Tamer Başar.
- Master of Applied Science in Electrical Engineering, University of Toronto, ON, Canada, Sept. 2006–June 2008. Master Thesis: An Investigation on Link Level End-to-End Power Control Models in WDM Optical Networks, advised by Prof. Lăcrășă Pavel.
- Bachelor of Engineering in Honors Electrical Engineering, McGill University, Montréal, QC, Canada, Sept. 2002–June 2006.

Current and Past Students

Academic Placements

- Ya-Ting Yang — Assistant Professor, National Taiwan University.
- Tao Li — Assistant Professor, City University of Hong Kong.
- Shutian Liu — Assistant Professor, City University of Hong Kong.
- Linan Huang — Assistant Professor, Tsinghua University.
- Juntao Chen — Associate Professor, Fordham University.
- Junaid Farooq — Associate Professor, University of Michigan.

Visiting Students

- Takuma Adams, University of Melbourne, Australia.

Current Postdoctoral Researcher

- Mohamed Arnouss (Fulbright Scholar)

Past Postdoctoral Researcher

- Song Fang (Oct. 2019–Aug. 2021)

Past Visiting Students

- Anh Tung Nguyen, Uppsala University, Sweden (Summer 2026)
- Yeongwoo Kim, Royal Institute of Technology (KTH), Sweden (Fall 2023)
- Kim Hammar, Royal Institute of Technology (KTH), Sweden (May 2023)

Graduated PhD Students

- Ya-Ting Yang, Ph.D. (2026). Assistant Professor, National Taiwan University. Dissertation: The Dialectics of Trust: Game-Theoretic Foundations for Trust in AI-Driven Socio-Technical Systems.
- Yunian Pan, Ph.D. (2026). Dissertation: The Variational Foundations of Multi-Agent Learning: A Unified Theory on Non-Equilibrium Learning Dynamics in Cyber-Physical Systems.
- Tao Li, Ph.D. (2024). Assistant Professor, City University of Hong Kong. Winner of the 2024 Dante Youla Award for Graduate Research Excellence.
- Shutian Liu, Ph.D. (2024). Assistant Professor, City University of Hong Kong. Dissertation: System-Scientific Foundations of Holonic Risk Analysis and Design in Socio-Cyber-Physical Systems.
- Yuhan Zhao, Ph.D. (2024). Dissertation: Learning and Game Theoretic Paradigms for Strategic Coordination in Multi-Agent Autonomous Systems.
- Yinan Hu, Ph.D. (2023). Dissertation: Advanced and Contemporary Applications of Classical and Quantum Detection Theories in Strategic Environments.
- Yunfei Ge, Ph.D. (2024). Dissertation: The Symbiosis of Trust and AI. Winner of the 2021 Chang Education Award; Winner of the 2024 Goodman Research Award.
- Tao Zhang, Ph.D. (2023). Dissertation: Dynamic Mechanism Design: From Theories to Applications. Winner of the 2020 Goodman Research Award.
- Linan Huang, Ph.D. (2022). Assistant Professor, Tsinghua University. Dissertation: Strategic Socio-Technical Defense Mechanisms for High-Confidence Cyber-Physical Systems. Winner of the 2022 Youla Award; Winner of the 2023 NYU-Wide Dissertation Award.
- Yunhan Huang, Ph.D. (2022). Dissertation: Sequential Decision Making with Strategic Information Acquisition and Adversarial Manipulation.
- Juntao Chen, Ph.D. (2020). Associate Professor, Fordham University. Dissertation: Trustworthy Cyber-Physical Networks by Design. Winner of the 2019 Youla Award; Winner of the 2021 Hessel Dissertation Award.
- Junaid Farooq, Ph.D. (2020). Associate Professor, University of Michigan. Dissertation: Cyber-Physical Dynamic Decision Mechanisms for Large-Scale IoT Systems. Winner of the 2019 Papoulis Award; 2020 Youla Award; 2021 NYU Dissertation Award.
- Rui Zhang, Ph.D. (2020). Dissertation: Game-Theoretic Approaches to Cyber Data Risk Mitigation and Transfer. Winner of the 2016 Rosenthal Award.
- Jeff Pawlick, Ph.D. (2018). Dissertation: A System Science Perspective on Deception for Cybersecurity in the Internet of Things.
- Zhiheng Xu, Ph.D. (2018). Dissertation: Security and Resilience of Cyber-Physical Systems.

Supervised MS Theses

- Pengyi Jiang, M.S. (2026). Thesis: Semantic Cooperative Game: Semantic Value Allocation and Coalition Formation in Multi-Agent LLM Collaboration, Master of Science in Electrical Engineering, New York University Tandon School of Engineering.
- Shizhen Jia, M.S. (2024). Thesis: Interference Avoidance for FMCW Automotive Radars Using Non-Cooperative Game Theory. Now Ph.D. Student at NYU.
- Lan Shi, M.S. (2022). Thesis: Heterogeneous Robots Collaborative Applications with Asymmetric Reinforcement Learning. Now Ph.D. Student at Purdue University.

Graduated MS Students

- Katherine Zhuo, Summer M.S. Student (2026). Research: Agentic AI risks and insurance.
- Xinhong Xie, M.S. (2024). Research: LLMs for Evading Censorship. Now Ph.D. Student at Penn State University.
- Haozhe Lei, M.S. (2022). Research: Black-Box Attacks in Meta Reinforcement Learning. Now Ph.D. Student at NYU.

Past Undergraduate Students

- Minjun Kang (2025): Agentic AI for proactive cybersecurity
- Xuan Liu (2024): Prosocial Irrationality for LLM Agents.
- Haoyang Shang (2024): Cyber Operations Research Gym (CybORG).
- Junjie Chen (2024): LLMs for Network Security.
- Juan Guevara (2023): Competitive RL as Sequence Modeling.
- Josh Schrock (2022): Game Theory and DMD of Cognitive Systems.

Teaching

Regular Courses

- ECE-GY 9953 Advanced Projects
- ECE-GY 9900 Seminar in Electrical and Computer Engineering
- ECE-GY 6233 System Optimization Methods
- ECE-GY 8233 Optimal Control
- ECE-UY 3064 Feedback Control
- ECE-GY 5213 Introduction to System Engineering

Developed Courses

- ECE-GY 6263 Game Theory
- ECE-GY 9213 Special Topics: Game Theory
- ECE-GY 9223 Special Topics: Reinforcement Learning
- ECE-GY 9233 Special Topics: Control and Optimization of Cyber-Physical Human Networks
- ECE-GY 9253 Special Topics: System Foundations of Cyber-Physical System Security
- VIP-UY 2001 Vertically Integrated Projects I: Internet of Controlled Things, 2017-2020

Courses under planning

- Artificial Intelligence and Games (Undergraduate and Graduate Course)
- Systems Engineering for Complex Problem-Solving (Graduate Course)
- Resilience System Engineering and Science (Graduate Course)

Short Courses, Tutorials, and Workshops

1. Organizer, Human-Aware AI Agents for the Cyber Battlefield: From Human Models to Autonomous Defense, AAAI 2026 Summer Symposium Series, June 22–24, 2026, Dongguk University, Seoul, South Korea.
2. Organizer, Tutorial, 23rd IFAC World Congress (IFAC WC 2026), August 23–28, 2026, Busan, Republic of Korea.
3. General Co-Chair, Workshop on Resilience in Next Generation Network Management (NGResNet), in conjunction with IEEE/IFIP Network Operations and Management Symposium (NOMS 2026), Rome, Italy, May 18–22, 2026.
4. General Co-Chair, First Workshop on Resilience in Next-Generation Networks (NGResNet 2025), in conjunction with IEEE International Conference on Computer Communications (INFOCOM 2025), London, United Kingdom, May 19, 2025.
5. Workshop Chair and Co-Organizer, 1st–4th IEEE CNS Cyber Resilience Workshop (CRW), held with the IEEE Conference on Communications and Network Security (IEEE CNS), 2022–2025.
6. Organized Multiple Invited Sessions on CPS Security and Resilience, 2023 IEEE 62nd Conference on Decision and Control (CDC), December 13–15, 2023, Singapore.
7. Organized Workshop on Control, Game, and Learning Theory for Security and Privacy, 2023 IEEE 62nd Conference on Decision and Control (CDC), December 12, 2023, Singapore.
8. Organized Minisymposium on Recent Advances in Dynamic Games and Control Theory and Their Connection to Data Science, 10th International Congress on Industrial and Applied Mathematics (ICIAM 2023), Waseda University, Tokyo, Japan, August 20–25, 2023.
9. Organized Open Invited Track for Modeling and Design of Secure and Resilient Control Systems, IFAC World Congress 2023, July 9–14, Yokohama, Japan.
10. Organized Workshop on Interdisciplinary and Convergent Research Foundations for Cyber Deception, August 2–3, 2023, New York, NY, USA.
11. Organized Invited Session on Dynamic Games and Networked Systems, 2022 IEEE 61st Conference on Decision and Control (CDC), December 6–9, 2022, Cancun, Mexico.
12. Organized Invited Session on Learning and AI Methods for CPS Security, 2022 IEEE 61st Conference on Decision and Control (CDC), December 6–9, 2022, Cancun, Mexico.
13. Delivered Tutorial on IoT Supply Chain Security Risk Assessment and Mitigation: Methodologies and Computational Tools, Military Communications Conference (MILCOM), November 28–December 2, 2022, National Capital Region, USA.

Scholarships, Awards and Honors

Fellowships and Scholarships

- Scholar in Residence, Digital Futures, KTH, 2022
- INFORMS Koopman Prize, 2022
- NYU Provost’s Global Research Initiatives (GRI) Fellowship, 2022
- NSF CAREER Award, 2019–2024
- NYU Goddard Fellowship, 2018

Best Paper Awards

- Best Paper Award: Y. Ge and Q. Zhu, “MEGA-PT: A Meta-game Framework for Agile Penetration Testing,” In: Proceedings of the International Conference on Decision and Game Theory for Security (GameSec 2024), pp. 24–44, 2024. https://doi.org/10.1007/978-3-031-74835-6_2.
- Best Student Paper Award: J. Huang and Q. Zhu, “PenHeal: A Two-Stage LLM Framework for Automated Pentesting and Optimal Remediation,” In: Proceedings of the Workshop on Autonomous Cybersecurity (AutonomousCyber ’24), ACM SIGSAC Conference on Computer and Communications Security (CCS), pp. 11–22, 2024. <https://doi.org/10.1145/3689933.3690831>.
- Best Paper Award: Y. Pan and Q. Zhu, “On Poisoned Wardrop Equilibrium in Congestion Games,” In: F. Fang, H. Xu, Y. Hayel (eds.), Decision and Game Theory for Security (GameSec 2022), Lecture Notes in Computer Science, vol. 13727. Springer, Cham, 2023. https://doi.org/10.1007/978-3-031-26369-9_10.

Invited Talks

1. Keynote Speaker, LSU Symposium on Control, Learning, and Intelligent Systems, Louisiana State University, May 30, 2026.
2. Invited Speaker, “Coalition, Consensus, and Coordination: A Tale of Noncooperative and Cooperative Games in Socio-Technical Systems,” 12th Midwest Workshop on Control and Game Theory (MWCGT), Purdue University, West Lafayette, IN, April 25, 2026.
3. Invited Speaker, “The Role of Information in Dynamic Games and Multi-Agent Learning,” Industrial Engineering Seminar, University of Pittsburgh, Pittsburgh, PA, March 19, 2026.
4. Invited Visiting Professor, Hanyang University and National Seoul University, November 25-29, 2024.
5. Invited Speaker, 2024 INFORMS Analytics Conference, Orlando, FL, April 14-16, 2024.
6. Keynote Speaker, Georgia Tech Decision and Control Conference, February 23, 2024.
7. Invited Speaker, GE EDGE Symposium, September 19-21, 2024.
8. Invited Speaker, The Future of Cyber Deception ARO Workshop, CMU, Pittsburgh, PA, October 2-6, 2023.
9. Invited Speaker, Midwest Control and Game Theory Workshop, April 22-23, 2023.
10. Invited Participant, NSF Joint US-European Workshop on “Flexible Electric Grid Critical Infrastructure for Resilient Society,” Philadelphia, PA, April 21-22, 2023.
11. Invited Participant, Transatlantic US-Ireland-Northern Ireland Workshop on Collaborative IoT/CPS Cybersecurity Research, October 20-21, 2022, CON- NECT Center, Dublin, Ireland.
12. Invited Participant, US-European Workshop on “Grid at the Edge: Towards the Zero-Carbon Power Grid with Improved Visibility, Safety, and Reliability,” Split, Croatia, May 23-24, 2022.
13. Invited Participant, NSF-Sponsored Workshop on “Grid at the Edge: Towards the Zero-Carbon Power Grid with Improved Visibility, Safety, and Reliability,” March 15 and March 22, 2022.
14. Invited Participant, “Facilitating U.S.-India Bilateral Research Collaborations Workshop,” NSF, March 15 and March 22, 2022.
15. Invited Participant, “URoL Workshop on Leveraging AI and Data Science for Predicting Mechanisms,” NSF, April 26, 2022.

Services

School Services

- Initiated Tandon-wide activities to prevent and respond to faculty Bullying and Other Demeaning and disruptive behavior (BORDER).
- Served as Editor-in-Chief (EiC) of Cyber Byte Newsletter of Center of Cyber Security (CCS), April 2021-Present.
- Faculty Engineers in Residence (F-EIR) to bridge the gap between academia and entrepreneurship.
- Invited to serve on PhD Dissertation Committee of
- Juan Herrera, PhD student from NYU Steinhardt
- Wilmer Cantos, PhD student from Department of Civil and Urban Engineering, NYU
- Karel Durkota, PhD Student from Czech Technical University in Prague, Czech Republic
- Sharad Kumar Singh, PhD Student from India Institute of Technology, Madras, India
- M.Venkateswararao.K, PhD Student from India Institute of Technology, Bombay, India
- Arnold Kouam Kounchou, PhD Student from University of Avignon, France
- Bo Pang, PhD Student at ECE NYU
- Leilei Cui, PhD Student at ECE NYU
- Mengzhe Huang, PhD Student at ECE NYU

University Services

- Served as the NYU representative of Institute for Information Infrastructure Protection (I3P).
- Faculty advisor for newly established NYU International Council on Systems Engineering (INCOSE) Student Chapter:<https://wp.nyu.edu/nyuincose/>.
- Developing Cooperative Research And Development Agreement (CRADA) between NYU and Army Research Laboratory (ARL).
- Initiated NYU collaboration with University of Strathclyde, Glasgow, UK on power systems research.

Editorship

- Guest Editor, IEEE Transactions on Industrial Cyber-Physical Systems Special Collection on Co-design of Cognition, Control, Computing and Communication for Industrial Cyber-Physical Systems, 2026.
- Guest Editor, IEEE Open Journal of the Communications Society Special issue on Trustworthy AI-Enabled Edge Computing in Next-Generation Wireless Networks, 2024
- Guest Editor, IEEE Journal on Selected Areas in Communications (JSAC) Special Issue on Co-Design of Communication, Computing, and Control in Industrial Cyber-Physical Systems, 2024.
- Guest Editor, IEEE Network Magazine Special Issue on Network-native Security, Safety, and Resilience Solutions for Critical Automation Systems Hosted in Cloud and Fog Computing, 2024.
- Guest Editor, IEEE Transactions on Control Systems Technology (TCST) Special Issue on Resilient Control of Cyber Physical Energy Systems, 2023.
- Associate Editor, IEEE Transactions on Aerospace and Electronic Systems, 2022- Present.
- Associate Editor, IEEE Transactions on Network Science and Engineering, 2024- Present.
- Associate Editor, IEEE Transactions on Network and Service Management (TNSM), Present.
- Associate Editor, Dynamic Games and Applications (DGAA), Elsevier, 2021- Present.

- Associate Editor in the area of Management of Energy Demand, Storage, and EVs, IET Transaction on Smart Grid, 2019-Present.

Scientific Committees and Activities

- Program Committee Member, Ninth AAAI/ACM Conference on AI, Ethics, and Society (AIES 2026), Malmö, Sweden, October 12–14, 2026.
- Member, Executive Board, The International Society of Dynamic Games (ISDG).
- Member, International Program Committee, 21st International Symposium on Dynamic Games and Applications (ISDG), July 20–23, 2026.
- TPC Member, 13th International Conference on Networks, Games, Control and Optimization (NETGCOOP), Palermo, Italy, October 14–16, 2026.
- Initiated NYU–TAU Cyber Collaboration on Human-Centric Cyber Resilience, May 2025.
- Board Member, INFORMS Military and Security Society (MAS), 2025.
- Chair, 15th Conference on Decision and Game Theory for Security (GameSec), New York, NY, October 16-18, 2024.
- TPC Member, Tech4Good: Economic and Computational Advances in Distributed Systems, Co-located with ICDCS 2024, Jersey City, NJ, June 16, 2024.
- TPC Member, 23rd Workshop on the Economics of Information Security, Dallas, TX, April 8-10, 2024.
- Organizer, ARO Workshop on Interdisciplinary Cyber Deception, New York, NY, August 17-18, 2023.
- Advisor, Inaugural NYC Intelligent Transportation System Workshop: Network Designs in the Era of Digital Mobility, NYU, January 16, 2024.
- Advisor, Inaugural NYC Cybersecurity Day Workshop: Human Elements in Cybersecurity, NYU, November 20, 2023.
- Advisory Panel Member, Engineering Higher Education, collaborated with CollegeBoard to develop curriculum outlines for the Advanced Placement (AP) Program in Engineering.
- Member, Technical Program Committees:
- IEEE International Conference on Computer Communications (INFOCOM), 2026.
- IEEE Global Communications Conference (GLOBECOM), Selected Areas in Communications Symposium: Smart Grid Communications and Control, Macau, China, December 7–11, 2026.
- Conference on New Innovations in Material Science (NIMS), 2026.
- International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc), 2026.
- International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt), 2026.
- IEEE International Conference on Computer Communications (INFOCOM), 2025.
- IEEE/IFIP Network Operations and Management Symposium (NOMS), 2025.
- International Symposium on Theory, Algorithmic Foundations, and Protocol Design for Mobile Networks and Mobile Computing (MobiHoc), 2025.
- Conference on Decision and Game Theory for Security (GameSec), 2015–2024.
- International Conference on Performance Evaluation Methodologies and Tools (VALUETOOLS), 2015–2024.
- International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks (WiOpt), 2015–2024.

NSF Panel Services

- Served on NSF ECCS EPCL Panel, 2026.
- Served on NSF CPS-FR Panel, 2026.
- Served on NSF CPS CAREER Panel, 2025.
- Served on NSF CMMI DCSD Panel, 2024.
- Served on NSF CPS Medium Panel, 2023.
- Served on NSF LEAP-HI Panel, 2022.
- Served on NSF EPCN CAREER Panel, 2022.
- Served on NSF CPS CAREER Panel, 2022.